IGCSE 0607 Practice Questions — E9.3 — Combined Questions & Markscheme

Probability of combined events

Total Questions: 9 — **Total Marks:** 60 Marks

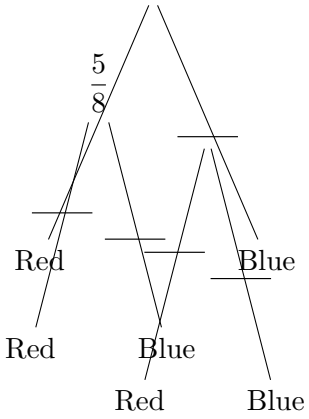
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  → **Paper 2 (Non-Calculator)** *(5 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

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1 A bag contains 5 red counters and 3 blue counters.
A counter is taken at random from the bag, its colour is noted and it is replaced.
A second counter is then taken at random.

(a) Complete the tree diagram.



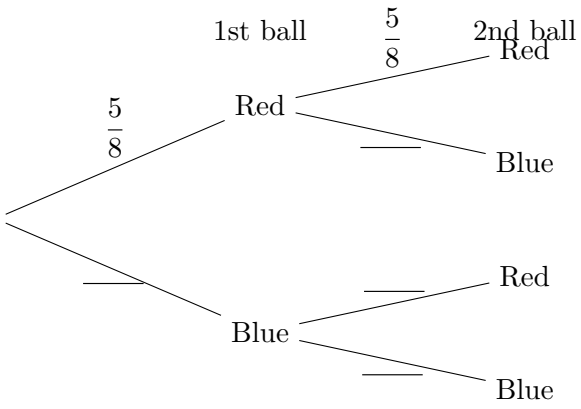
[2]

(b) Calculate the probability that both counters are red.

..... **[2]**

2 A bag contains 8 balls.
5 of the balls are Red and 3 of the balls are Blue.
A ball is taken at random from the bag, its colour is noted, and it is replaced.
A second ball is then taken at random from the bag.

(a) Complete the tree diagram.



[2]

(b) Calculate the probability that both balls are Red.

..... **[1]**

(c) Calculate the probability of picking one Red ball and one Blue ball in any order.

..... **[2]**

3 A box contains n counters.
3 of the counters are Red and all the remaining counters are Blue.
Two counters are taken at random from the box, without replacement.
The probability that both counters are Red is $\frac{1}{15}$.

(a) Show that $n^2 - n - 90 = 0$.

[3]

(b) Find the value of n.

$n =$ ..... **[2]**

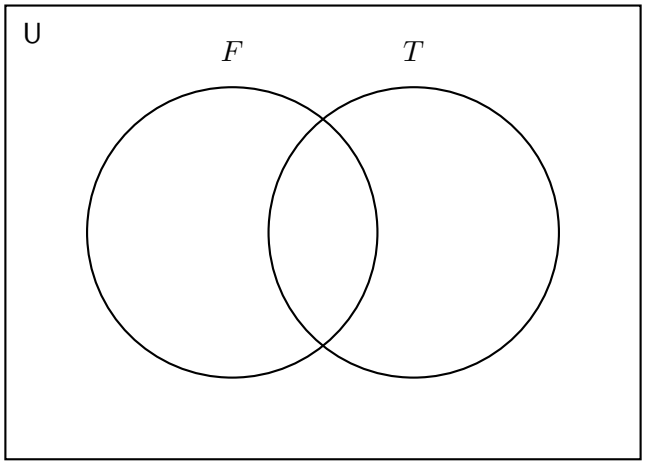
(c) Calculate the probability that exactly one of the two counters chosen is Red.

..... **[3]**

4 There are 30 students in a class.
U = {students in the class}
F = {students who play Football}
T = {students who play Tennis

18 students play Football. 14 students play Tennis. 4 students do not play Football and do not play Tennis.

(a) Complete the Venn diagram to show the number of students in each region.



[2]

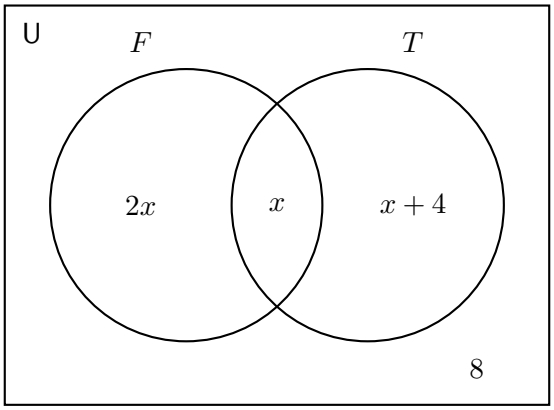
(b) A student is chosen at random from the class.
Find $P(F \cap T)$.

$P(F \cap T) =$ ..... **[1]**

(c) Two students are chosen at random from the class.
Calculate the probability that both students play Football but not Tennis.

..... **[3]**

5 A group of 40 students are asked if they play Football (F) or Tennis (T).
The Venn diagram shows the number of students in each region in terms of x.



(a) Find the value of x.

$x =$ ..... **[2]**

(b) Find $P(F \cup T)$.

..... **[1]**

(c) Two students are chosen at random from the group of 40.
Calculate the probability that both students play Football only.

..... **[3]**

6 A bag contains 4 red discs and some blue discs.
Let the total number of discs in the bag be x.
Two discs are taken at random from the bag, without replacement.
The probability that both discs are red is $\frac{1}{11}$.

(a) Show that $x^2 - x - 132 = 0$.

[3]

(b) Find the number of blue discs in the bag.

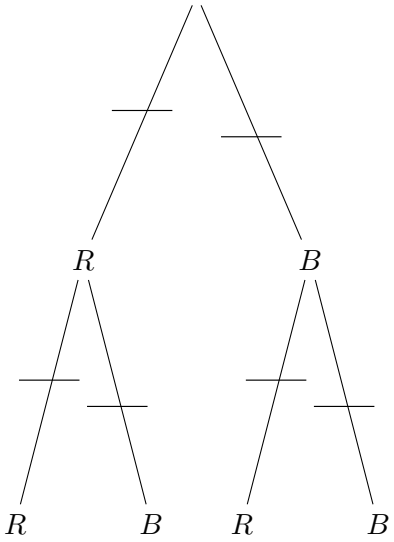
..... **[3]**

(c) Calculate the probability that one disc is red and one disc is blue.

..... **[3]**

7 A bag contains 7 red counters and 5 blue counters.
A counter is taken at random, its colour is noted, and it is NOT replaced in the bag.
A second counter is then taken at random and its colour is noted.

(a) Complete the tree diagram by writing the missing probabilities on the branches.



[3]

(b) Calculate the probability that both counters are red.

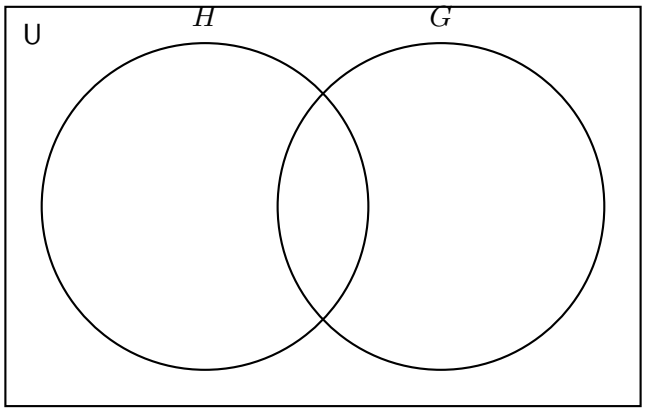
..... **[2]**

(c) Calculate the probability that one counter is red and one counter is blue.

..... **[2]**

8 In a group of 50 students, 32 study History (H), 21 study Geography (G), and 6 study neither subject.

(a) Complete the Venn diagram to show the number of students in each region.



[3]

(b) A student is chosen at random from the group of 50.

Find:

(i) $P(H \cap G)$

$P(H \cap G) =$ ..... **[1]**

(ii) $P(H \cup G')$

$P(H \cup G') =$ ..... **[1]**

(c) Two students who study History are chosen at random.
Calculate the probability that both students also study Geography.

..... **[2]**

9 A box contains x red balls and 4 blue balls.
Two balls are taken from the box at random, without replacement.
The probability that both balls are red is $\frac{1}{3}$.

(a) Show that $x^2 - 5x - 6 = 0$.

[3]

(b) Work out the number of red balls in the box.

..... **[2]**

(c) Calculate the probability that the two balls taken have different colours.

..... **[3]**

1 **(a)** $\frac{3}{8}$ correctly placed on the first branch for Blue **[B1]**

$\frac{5}{8}$ and $\frac{3}{8}$ correctly placed on all second branches **[B1]**

(b) $\frac{5}{8} \times \frac{5}{8}$ **[M1]**

$\frac{25}{64}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** $\frac{3}{8}$ correctly placed for 1st ball Blue **[B1]**

$\frac{3}{8}$, $\frac{5}{8}$, and $\frac{3}{8}$ correctly placed on all three remaining 2nd ball branches **[B1]**

(b) $\frac{25}{64}$ **[B1]**

(c) $\left(\frac{5}{8} \times \frac{3}{8}\right) + \left(\frac{3}{8} \times \frac{5}{8}\right)$ **[M1]**

$\frac{30}{64}$ or $\frac{15}{32}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** $\frac{3}{n} \times \frac{2}{n-1} = \frac{1}{15}$ **[M1]**

$\frac{6}{n(n-1)} = \frac{1}{15}$ **[M1]**

$n^2 - n = 90 \implies n^2 - n - 90 = 0$ **[A1]**

(b) $(n-10)(n+9) = 0$ **[M1]**

$n = 10$ **[A1]**

(c) Number of Blue counters $= 7$ **[B1]**

$\left(\frac{3}{10} \times \frac{7}{9}\right) + \left(\frac{7}{10} \times \frac{3}{9}\right)$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

4 **(a)** 6 placed correctly in the intersection **[B1]**

12 in F only, 8 in T only, and 4 outside the circles **[B1]**

(b) $\frac{6}{30}$ or $\frac{1}{5}$ or 0.2 **[B1]**

(c) $\frac{12}{30}$ seen as the probability for the first student **[M1]**

$\frac{12}{30} \times \frac{11}{29}$ **[M1]**

$\frac{132}{870}$ or $\frac{22}{145}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

5 **(a)** $2x + x + (x + 4) + 8 = 40$ **[M1]**

$4x + 12 = 40 \implies x = 7$ **[A1]**

(b) $\frac{32}{40}$ or $\frac{4}{5}$ **[B1]**

(c) Number of students playing Football only $= 14$ **[B1]**

$\frac{14}{40} \times \frac{13}{39}$ **[M1]**

$\frac{182}{1560}$ or $\frac{7}{60}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** $\frac{4}{x} \times \frac{3}{x-1}$ **[M1]**

$\frac{12}{x(x-1)} = \frac{1}{11}$ or $12 \times 11 = x(x-1)$ **[M1]**

$132 = x^2 - x \implies x^2 - x - 132 = 0$ **[A1]**

(b) $(x - 12)(x + 11) = 0$ or correct substitution into the quadratic formula **[M1]**

$x = 12$ **[A1]**

Number of blue discs $= 12 - 4 = 8$ **[A1]**

(c) $\frac{4}{12} \times \frac{8}{11}$ or $\frac{32}{132}$ **[M1]**

$\left(\frac{4}{12} \times \frac{8}{11}\right) \times 2$ **[M1]**

$\frac{64}{132}$ or $\frac{16}{33}$ **[A1]**

[Back to Question] *[Total Marks: 9]*

7 **(a)** First stage branches correctly labeled: $\frac{7}{12}$ for R and $\frac{5}{12}$ for B **[B1]**

Second stage branches from R correctly labeled: $\frac{6}{11}$ for R and $\frac{5}{11}$ for B **[B1]**

Second stage branches from B correctly labeled: $\frac{7}{11}$ for R and $\frac{4}{11}$ for B **[B1]**

(b) Multiplies probabilities along the R, R path: $\frac{7}{12} \times \frac{6}{11}$ **[M1]**

$\frac{42}{132}$ or $\frac{7}{22}$ **[A1]**

(c) Identifies and sums both correct combinations: $\left(\frac{7}{12} \times \frac{5}{11}\right) + \left(\frac{5}{12} \times \frac{7}{11}\right)$ or $2 \times \left(\frac{35}{132}\right)$ **[M1]**

$\frac{70}{132}$ or $\frac{35}{66}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

8 (a) Places 6 outside both circles **[B1]**

Calculates intersection: $32 + 21 + 6 - 50 = 9$ and places 9 in $H \cap G$ **[B1]**

Places 23 in H only and 12 in G only **[B1]**

(b) (i) $\frac{9}{50}$ **[B1]**

(ii) $\frac{38}{50}$ or $\frac{19}{25}$ **[B1]**

(c) Recognises the reduced sample space and uses probability without replacement: $\frac{9}{32} \times \frac{8}{31}$ **[M1]**

$\frac{72}{992}$ or $\frac{9}{124}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

9 (a) Sets up correct equation for probability of two red balls: $\frac{x}{x+4} \times \frac{x-1}{x+3} = \frac{1}{3}$ **[M1]**

Cross-multiplies correctly to clear fractions: $3x(x-1) = (x+4)(x+3)$ **[M1]**

Expands and rearranges correctly with no errors:
$3x^2 - 3x = x^2 + 7x + 12 \implies 2x^2 - 10x - 12 = 0 \implies x^2 - 5x - 6 = 0$ **[A1]**

(b) Factorises or solves the quadratic equation: $(x-6)(x+1) = 0$ **[M1]**

6 (and explicitly or implicitly rejects $x = -1$) **[A1]**

(c) Uses $x = 6$ to determine probabilities for one red and one blue ball: $\frac{6}{10} \times \frac{4}{9}$ **[M1]**

Accounts for both (R, B) and (B, R) combinations: $2 \times \left(\frac{6}{10} \times \frac{4}{9}\right)$ or $\frac{24}{90} + \frac{24}{90}$ **[M1]**

$\frac{48}{90}$ or $\frac{8}{15}$ **[A1]**

[Back to Question] *[Total Marks: 8]*